DAV PUBLIC SCHOOL, BSEB COLONY, PATNA-23

SUMMER VACATION HHW FOR STD. X

POLYNOMIALS

1. **Find the zeroes of the following quadratic polynomials and verify the relationship between the zeroes and their coefficients:**
   (i) $x^2 - 2x - 8$ (ii) $4s^2 - 4s + 1$
2. **Divide the polynomial p(x) by the polynomial g(x) and find the quotient and remainder in each of the following and also verify the division algorithm:**
   (i) $p(x) = x^3 - 3x^2 + 5x - 3$, $g(x) = x^2 - 2$ (ii) $p(x) = x^4 - 3x^2 + 4x + 5$, $g(x) = x^2 + 1 - x$
3. **Obtain all other zeroes of $3x^4 + 6x^3 - 2x^2 - 10x - 5$, if two of its zeroes are $\sqrt{\{5/3\}}$ and - $\sqrt{\{5/3\}}$**
4. **If the zeroes of the polynomial $x^3 - 3x^2 + x + 1$ are a-b, a, a + b, find a and b.**
5. **If two zeroes of the polynomial $x^4 - 6x^3 - 26x^2 + 138x - 35$ are $2 \pm \sqrt{3}$, find other zeroes,**
6. **Does the polynomial $a^4 + 4a^2 + 5$ have real zeroes?**
7. **Find the quadratic polynomial if its zeroes are 0, $\sqrt{3}$.**
8. **α and β are zeroes of the quadratic polynomial $x^2 - 6x + y$. Find the value of 'y' if $3\alpha + 2\beta = 20$**
9. **If 4 is a zero of the cubic polynomial $x^3 - 3x^2 - 10x + 24$, find its other two zeroes.**
10. **Find the zeroes of the quadratic polynomial $7y^2 - (11/3)y - (2/3)$ and verify the relationship between the zeroes and the coefficients.**
11. **Find all zeroes of the polynomial $(2x^4 - 9x^3 + 5x^2 + 3x - 1)$ if two of its zeroes are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$.**
12. **If the remainder on division of $x^3 + 2x^2 + k x + 3$ by $x - 3$ is 21, find the quotient and the value of k. Hence, find the zeroes of the cubic polynomial $x^3 + 2x^2 + k x - 18$.**
13. **For which values of a and b, are the zeroes of $q(x) = x^3 + 2x^2 + a$ also the zeroes of the polynomial $p(x) = x^5 - x^4 - 4x^3 + 3x^2 + 3x + b$? Which zeroes of p(x) are not the zeroes of q(x)?**
14. **Find a quadratic polynomial, the sum and product of whose zeroes are $\sqrt{2}$ and -3/2, respectively. Also find its zeroes**

15.If α and β are the zeroes of the polynomial $ax^2 + bx + c$, find the value of i) $\alpha^2 + \beta^2$. ii) $\alpha^3 + \beta^3$ iii) $\alpha^4 + \beta^4$ iv) Sum of reciprocal of zeroes

16. Find the condition that zeroes of polynomial $p(x) = ax^2 + bx + c$ are reciprocal of each other

17. If the zeroes of the polynomial $x^2 + px + q$ are double in value to the zeroes of $2x^2 - 5x - 3$, find the value of p and q

18. If α and β are zeroes of $p(x) = kx^2 + 4x + 4$, such that $\alpha^2 + \beta^2 = 24$, find k

19. Find the values of a and b so that $x^4 + x^3 + 8x^2 + ax - b$ is divisible by $x^2 + 1$

20) If α, β are the zeroes of the polynomial $2x^2 + 5x + k$ such that $\alpha^2 + \beta^2 + \alpha\beta = 21/4$, then find k.

REAL NUMBERS

1. **Find the HCF of the smallest prime number and the smallest composite number.**
2. **If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then find lcm of a and b.**
3. **Explain why 7 * 11 * 13 + 13 is a composite number.**
4. **Check whether 6^n can end with the digit 0 for any natural number n.**
5. **Find the HCF and LCM of 12, 15 and 21 using the prime factorisation method.**

6. **If the HCF of 65 and 117 is expressible in the form 65n - 117, find the value of n.**
7. **Prove that $\sqrt{3}$ is an irrational number.**
8. **Prove that $\sqrt{3} + 2$ is an irrational number, given that $\sqrt{3}$ is irrational.**
9. **Find the largest number that divides 70 and 125, leaving remainders 5 and 8 respectively.**
10. **Two numbers are in the ratio 2:3 and their LCM is 180. Find the HCF of these numbers.**
11. **Given that HCF(306, 657) = 9, find the LCM(306, 657).**
12. **Prove that $\sqrt{5} - \sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is irrational.**
13. **Three bells toll together at intervals of 9, 12, 15 minutes respectively. If they start tolling together, after what time will they next toll together?**

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